Roll No. | | | | | | | | | | | | | Total No. of Pages : 02

Total No. of Questions : 09

B.Tech (Computer Engg.) (Sem.–7)

NETWORK SECURITY AND CRYPTOGRAPHY

Subject Code : BTCS701-18

M.Code : 92906

Date of Examination : 13-12-22

Time : 3 Hrs. Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.
3. SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.

SECTION-A

1. Write briefly :

a) Discuss three security goals.

b) Distinguish between stream and block ciphers.

c) What is honeypots?

d) What is triple DES? List any one drawback of it.

e) State two Strategies for testing primality.

f) Describe the idea of Merkel-Damgard scheme in cryptographic hash functions.

g) Define a trapdoor one way function in asymmetric cryptography.

h) Justify metamorphism is body polymorphic in viruses attack.

i) How Digital Signature scheme is used in security?

j) Give the attacks possible on Diffie-Hellman algorithm.

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SECTION–B

2. What is cryptanalysis? Explain any three ways of designing cryptanalysis attacks.

3. Discuss the principle of RSA algorithm. Give its flow diagram.

4. Distinguish between HMAC and CMAC.

5. Explain how malicious programs exploit the principle of stack overflow in attacking system.

6. Give Simple protocol for key distribution centers and its drawbacks.

SECTION-C

7. List all three types of IDPS and differentiate between them.

8. List properties of Cryptographic functions. Using any one example explain how they follow them.

9. Compare and contrast AES and DES on basis of : Substitution, permutation, size of block, size of key.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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